The Journal of Open Source Software

Orbital Dynamics in X-RAY stellar binary systems

Graciela Sanjurjo-Ferrín[1*], **Jessica Planelles Villalva**[1*], **Jose Miguel Torrejón**[1*], **and Jose Joaquín Rodes-Roca**[1*]

1 Instituto Universitario de Física Aplicada a las Ciencias y las Tecnologías, Universidad de Alicante, 03690 Alicante, Spain * These authors contributed equally.

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Warrick Ball

Reviewers:

- @matteobachetti
- @tddesjardins

Submitted: 19 June 2024
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

X-ray astronomy is a young discipline, spanning no more than a few decades. The Earth's atmosphere is opaque to this type of radiation, so observations in this part of the spectrum were delayed until the space era began, when rocket launchers carrying X-ray telescopes revealed the universe from a brand-new point of view.

X-ray binaries are tremendously interesting systems. In these extreme environments, a compact object—either a neutron star or black hole—draws in matter from a companion star, producing intense X-ray emissions. These systems offer a unique window into extreme physics, from the effects of strong gravity and relativistic jets to the presence of intense magnetic fields.

Orbital modulations are observed in nearly all X-ray binary systems. These variations arise from the orbital motions of the system, driven by the relative velocities of the two stars and their changing configurations with respect to each other and the observer.

To aid in the study of these modulations, we introduce **xraybinaryorbit**, a user-friendly Python package designed to simplify the analysis of orbital modulations in X-ray binaries. There are three groups of funtions within this package, theoretical, fitting and timing. The software provides a user-friendly interface for managing the various parameters that influence orbital modulations. Upon first use, the user inputs parameters through a form, which are then saved in a file within the running directory, this file is automatically loaded in future runs, eliminating the need to re-enter all parameters. The software offers two fitting approaches: least squares (LS) and particle swarm optimization (PSO) *PySwarms* (2022). The LS methods is faster but may fail to converge in certain cases, whereas the PS0 is more robust but computationally intensive. Some results obtained with this packages are shown in Figure 1.

There are many software that can be applied to X-ray binary system analysis, including **Stingray** (Huppenkothen et al., 2019) which focus on timing analysis of astronomical data, **Jaxspec** which specializes in spectral fitting using Bayesian inference. **Lightkurve**, facilitates the analysis of time-series data from space missions like Kepler and TESS. A well know comprehensive package for general astronomy computations and data handling is **Astropy** (Astropy Collaboration et al., 2022). In contrast **xraybinaryorbit** is highly specialized, providing detailed insights into orbital modulations within X-ray binary systems.

Science Behind

Conservation of Angular Momentum

In systems with non-zero eccentricity, the orbital phase varies non-linearly with observational time due to increased speed at periastron, governed by the conservation of angular momentum. Kepler's laws further explain that objects sweep out equal areas in equal times (Carroll & Ostlie, 2006).

$$r^2 \cdot \omega = Area$$

We account for this in all functions and provide tools to convert phase to time and vice versa.

CAK Model

The CAK model (Castor et al., 1975) describes radiation-driven winds in massive stars, with wind velocity, density, and ionization state varying with distance to the companion star.

$$\rho = \frac{\dot{M}}{4\pi v R^2}$$

Here, ρ is wind density, $\dot{M}$ is the mass loss rate, v is the wind velocity and R is the distance to the companion. We assume a spherical, smooth and un-ionized wind distribution.

Accretion Luminosity and Ionization

Accretion powers many close binary systems (Frank et al., 2002). The accretion luminosity is given by:

$$L_{ac} = \frac{GM\dot{M}}{R}$$

where L_{ac} is the luminosity, G is the gravitational constant, M is the compact object mass, and R is the compact object radius. The ionization parameter ξ is defined as:

$$\xi = \frac{L_{\mathrm{X}}}{n(r_{\mathrm{X}})r_{\mathrm{X}}^2}$$

Doppler Effect

The Doppler effect refers to the change in frequency or wavelength of a wave in relation to an observer moving relative to the source.

The Doppler velocity provoked by orbital motion can be calculated as:

$$v_D = (-r\omega \sin\phi \sin i)$$

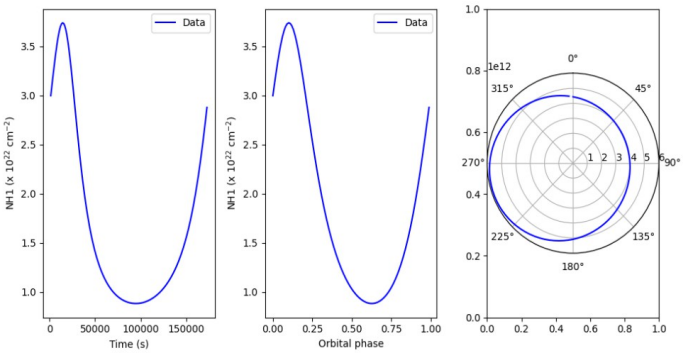
The observed Doppler length wave can be written as:

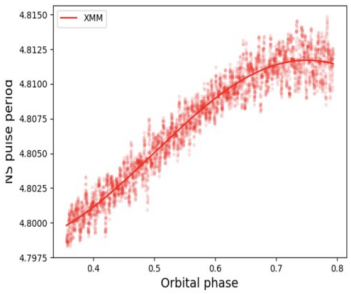
$$\lambda_D = \lambda_{\text{rest}} \left(1 + \frac{v_D}{c}\right)$$

where r is the orbital radius, ω is angular velocity, i is inclination, and λ_{D} and λ_{rest} are the Doppler-shifted and rest wavelengths, respectively.

DRAFT

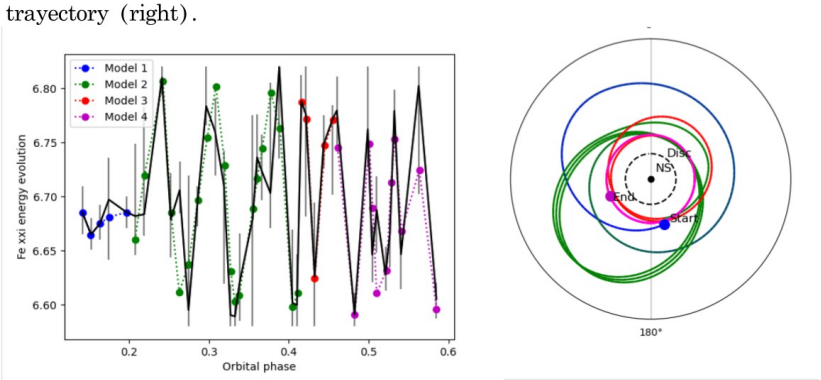
Theoretical absorption column (NH) faced by the emitted radiation depending on the emitted orbital phase. From left to right vs time, orbital phase and the pole-on orbital representation.





Neutron star spin period evolution as it travels around the orbit. The red solid line represent the fit. With it, we were able to recover and constrain the orbital parameters.

Fe XXV Doppler shifts fitted with Particle swarm (left) to retrieve the emitting plasma trayectory (right).



Real Ionization maps of the orbital's plane of a source through different observations.

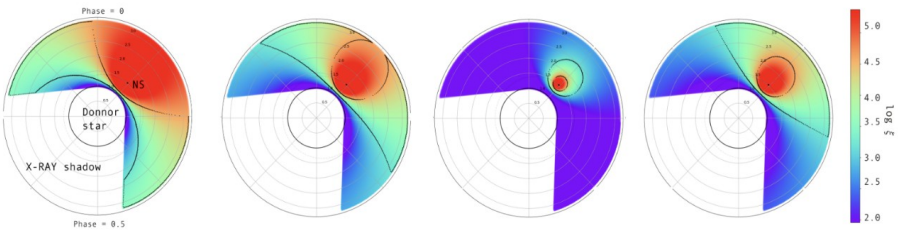


Figure 1: Some results obtained with the functions contained in this package.

Statement of Need

Within our observations, we might detect slight oscillating shifts in the center of emission lines and/or small variations in a neutron star's spin period. This phenomenon is likely caused by

the Doppler effect. Our code helps translate this data into the orbital parameters responsible for such Doppler shifts.

However, this seemingly simple process becomes complex when accounting for the several factors involved: inclination, eccentricity, periapsis, and distance to the barycenter (which depend on the masses of the stars involved). If the eccentricity is greater than zero, the velocity around the orbit is not constant, making the analysis even more intricate.

When considering stellar wind (the matter accreted by compact objects), various combinations of these factors result in orbital modulations. Eccentricity causes changes in wind density throughout the orbit, affecting the amount of accreted matter. Both eccentricity and inclination influence the absorption column faced by the emitted radiation and the ionization of the stellar wind, which also varies with orbital phase.

Although these orbital modulations are conceptually simple, they are challenging to analyze. At the same time, they can provide tremendous insights into the orbital mechanics and wind properties of our systems, may help to complete the puzzle when we already have some pieces of information or to give us a list of possible scenarios compatible with our data. This is where our tools provide valuable support.

Historically, the primary limitation in this type of analysis has been the lack of resolution for detailed phase-resolved observations. However, upcoming high-resolution missions like XRISM (XRISM Science Team, 2022) and New Athena [2016SPIE.9905E..2FB] promise to significantly improve the quality of these analyses. In addition to better resolution, advances in computational power have been crucial. Many of these tools have already been successfully applied to studies using XMM-Newton and Chandra data, enabling analyses that were previously impossible. (Sanjurjo-Ferrín, G. et al., 2024; Sanjurjo-Ferrín et al., 2021; **?**)

Acknowledgements

This research has been funded by the ASFAE/2022/02 project from the Generalitat Valenciana.

References

Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., Earl, N., Starkman, N., Bradley, L., Shupe, D. L., Patil, A. A., Corrales, L., Brasseur, C. E., Nothe, M., Donath, A., Tollerud, E., Morris, B. M., Ginsburg, A., Vaher, E., Weaver, B. A., Tocknell, J., Jamieson, W., … Astropy Project Contributors. (2022). The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package. *935*(2), 167. https://doi.org/10.3847/1538-4357/ac7c74

Bonyadi, M. R., & Michalewicz, Z. (2017). Particle Swarm Optimization for Single Objective Continuous Space Problems: A Review. *Evolutionary Computation*, *25*(1), 1–54. https://doi.org/10.1162/EVCO_r_00180

Carroll, B. W., & Ostlie, D. A. (2006). *An introduction to modern astrophysics and cosmology*.

Castor, J. I., Abbott, D. C., & Klein, R. I. (1975). Radiation-driven winds in Of stars. *195*, 157–174. https://doi.org/10.1086/153315

Frank, J., King, A., & Raine, D. (2002). *Accretion power in astrophysics* (3rd ed.). Cambridge University Press.

Huppenkothen, D., Bachetti, M., Stevens, A. L., Migliari, S., Balm, P., Hammad, O., Khan, U. M., Mishra, H., Rashid, H., Sharma, S., Martinez Ribeiro, E., & Valles Blanco, R. (2019). Stingray: A Modern Python Library for Spectral Timing. *881*(1), 39. https://doi.org/10.3847/1538-4357/ab258d

DRAFT

PySwarms: A python-based swarm optimization library. (2022). https://pythonhosted.org/pyswarm/.

Sanjurjo-Ferrín, G., Torrejón, J. M., Oskinova, L., Postnov, K., Rodes-Roca, J. J., Schulz, N., & Nowak, M. (2024). Revealing the state transition of cen x-3 at high spectral resolution with chandra. *A&A*, *690*, A360. https://doi.org/10.1051/0004-6361/202449724

Sanjurjo-Ferrín, G., Torrejón, J. M., Postnov, K., Oskinova, L., Rodes-Roca, J. J., & Bernabeu, G. (2021). X-ray variability of the HMXB Cen X-3: evidence for inhomogeneous accretion flows. *501*(4), 5892–5909. https://doi.org/10.1093/mnras/staa3953

XRISM Science Team. (2022). XRISM Quick Reference. *arXiv e-Prints*, arXiv:2202.05399. https://doi.org/10.48550/arXiv.2202.05399

DRAFT